1. Lesson 5: Fractional Sobolev space

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Let Ω be an open set in $\mathbb{R}^n$. We fix $s \in (0,1)$. For any $p \in [1,+\infty)$ we define the fractional Sobolev space $W^{s,p}(\Omega)$ as follows:

$$W^{s,p}(\Omega) := \left\{ u \in L^p(\Omega) \,:\, [u]_{W^{s,p}(\Omega)} < \infty \right\}; \tag{1.1}$$

where

$$[u]_{W^{s,p}(\Omega)} := \left(\int_\Omega \int_\Omega \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}} dxdy \right)^{\frac{1}{p}},$$

is the so-called Gagliardo (semi)-norm of u. The space $W^{s,p}(\Omega)$ is endowed with the natural norm

$$\|u\|_{W^{s,p}(\Omega)} := \left(\int_\Omega |u(x)|^p dx + \int_\Omega \int_\Omega \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}} dxdy \right)^{\frac{1}{p}}. \tag{1.2}$$

To show the properties of the fractional Sobolev space we need the following:

$$\int_{B_1} \frac{1}{|x|^\alpha} dx = \begin{cases} \text{convergent} & \alpha < n \\ \text{divergent} & \alpha \geq n \end{cases} \tag{1.3}$$

$$\int_{\mathbb{R}^n \setminus B_1} \frac{1}{|x|^\alpha} dx = \begin{cases} \text{convergent} & \alpha > n \\ \text{divergent} & \alpha \leq n, \end{cases} \tag{1.4}$$

where B_1 is the unit ball centred in zero. It is worth noticing that, as in the classical case, the space $W^{s',p}$ is continuously embedded in $W^{s,p}$ when $s' < s$, as next result points out.

Proposition 1.1. *Let $p \in [1,+\infty)$ and $0 < s \leq s' < 1$. Let Ω be an open set in $\mathbb{R}^n$ and $u : \Omega \to \mathbb{R}^n$ be a measurable function. Then*

$$\|u\|_{W^{s,p}(\Omega)} \leq C \|u\|_{W^{s',p}(\Omega)}$$

for some suitable positive constant $C = C(n,s,p) \geq 1$. In particular

$$W^{s',p}(\Omega) \subseteq W^{s,p}(\Omega).$$

Proof. First

$$\begin{aligned} \int_\Omega \int_{\Omega \cap \{|x-y| \geq 1\}} \frac{|u(x)|^p}{|x-y|^{n+sp}} dxdy &\leq \int_\Omega \left(\int_{\{|z| \geq 1\}} \frac{1}{|z|^{n+sp}} dz \right) |u(x)|^p dx \\ &\leq C(n,s,p) \|u\|^p_{L^p(\Omega)}, \end{aligned}$$

where we used the fact that the kernel $1/|z|^{n+sp}$ is integrable since $n + sp > n$, see (1.4). Taking into account the above estimate and using the fact that $(a+b)^p \leq 2^{p-1}(a^p + b^p)$, with $a, b > 0$, it follows

$$\begin{aligned} \int_\Omega \int_{\Omega \cap \{|x-y| \geq 1\}} \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}} dxdy &\leq 2^{p-1} \int_\Omega \int_{\Omega \cap \{|x-y| \geq 1\}} \frac{|u(x)|^p + |u(y)|^p}{|x-y|^{n+sp}} dxdy \\ &\leq 2^p C(n,s,p) \|u\|^p_{L^p(\Omega)}. \end{aligned} \tag{1.5}$$

On the other hand since $s \leq s'$ we have

$$\int_\Omega \int_{\Omega \cap \{|x-y| < 1\}} \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}} dxdy \leq \int_\Omega \int_{\Omega \cap \{|x-y| < 1\}} \frac{|u(x)-u(y)|^p}{|x-y|^{n+s'p}} dxdy. \tag{1.6}$$

Thus by combining (1.5) and (1.6), we get

$$\begin{aligned}\int_\Omega\int_\Omega \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}}dxdy &= \int_\Omega\int_{\Omega\cap\{|x-y|\geq 1\}} \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}}dxdy + \int_\Omega\int_{\Omega\cap\{|x-y|<1\}} \frac{|u(x)-u(y)|^p}{|x-y|^{n+sp}}dxdy \\ &\leq 2^pC(n,s,p)\|u\|^p_{L^p(\mathbb{R}^n)} + \int_\Omega\int_\Omega \frac{|u(x)-u(y)|^p}{|x-y|^{n+s'p}}dxdy\end{aligned}$$

and so

$$\begin{aligned}\|u\|^p_{W^{s,p}(\Omega)} &\leq (2^pC(n,s,p)+1)\|u\|^p_{L^p(\mathbb{R}^n)} + \int_\Omega\int_\Omega \frac{|u(x)-u(y)|^p}{|x-y|^{n+s'p}}dxdy \\ &\leq C(n,s,p)\|u\|^p_{W^{s',p}(\Omega)},\end{aligned}$$

which gives the desired estimate, up ro relabeling the constant $C(n,p,s)$.

□

In this lecture, we focus on the case $p=2$. This is quite an important case since the fractional Sobolev spaces $W^{s,2}(\mathbb{R}^n)$ turn out to be Hilbert space. It is usually related to the fractional Laplacian operator $(-\Delta)^s$, where, for any $u\in\mathcal{S}(\mathbb{R}^n)$ and $s\in(0,1)$, $(-\Delta)^s$ is defined as

$$(-\Delta)^s u(x) = \frac{1}{2}C(n,s)\int_{\mathbb{R}^n}\frac{2u(x)-u(x+y)-u(x-y)}{|y|^{n+2s}}dy. \tag{1.7}$$

Here $C(n,s)$ is a dimensional constant that depends on n and s, precisely given by

$$C(n,s) := \left(\int_{\mathbb{R}^n}\frac{1-\cos(\xi_1)}{|\xi|^{n+2s}}d\xi\right)^{-1}. \tag{1.8}$$

Now, we are ready to take into account an alternative definition of the space $H^s(\mathbb{R}^n)=W^{s,2}(\mathbb{R}^n)$ via the Fourier transform. Precisley, we may define

$$\hat{H}^s(\mathbb{R}^n) := \left\{u\in L^2(\mathbb{R}^n)\ :\ \int_{\mathbb{R}^n}(1+|\xi|^{2s})|\mathcal{F}u(\xi)|^2d\xi<\infty\right\} \tag{1.9}$$

and we observe that the above definition, unlike the ones via the Gagliardo norm in (1.2), is valid also for any $s\geq 1$.
The goal of this lecture is to show the equivalence of the space $\hat{H}^s(\mathbb{R}^n)$ defined in (1.9) with the one defined in (1.1) (with $p=2$).

Proposition 1.2. *Let $s\in(0,1)$ and let $(-\Delta)^s:\mathcal{S}(\mathbb{R}^n)\to L^2(\mathbb{R}^n)$ be the fractional Laplacian operator defined by (1.7). Then for $u\in\mathcal{S}(\mathbb{R}^n)$ we have*

$$(-\Delta)^s u(x) = \mathcal{F}^{-1}((2\pi|\xi|)^{2s}\hat{u}(\xi)).$$

Proof. Considering (1.7) and apply the Fourier transform [1], and using the Euler identity we obtain

$$\begin{aligned}\mathcal{F}\left((-\Delta)^s u(x)\right) &= \frac{C(n,s)}{2}\int_{\mathbb{R}^n}\frac{\mathcal{F}\left(2u(x)-u(x+y)-u(x-y)\right)}{|y|^{n+2s}}dy \\ &= \frac{C(n,s)}{2}\int_{\mathbb{R}^n}\hat{u}(\xi)\frac{2-e^{2\pi i\xi\cdot y}-e^{-2\pi i\xi\cdot y}}{|y|^{n+2s}}dy \\ &= \frac{C(n,s)}{2}\int_{\mathbb{R}^n}\hat{u}(\xi)\frac{2-\cos(2\pi\xi\cdot y)-i\sin(2\pi\xi\cdot y)-\cos(2\pi\xi\cdot y)+i\sin(2\pi\xi\cdot y)}{|y|^{n+2s}}dy \\ &= C(n,s)\hat{u}(\xi)\int_{\mathbb{R}^n}\frac{1-\cos(2\pi\xi\cdot y)}{|y|^{n+2s}}dy. \end{aligned}\tag{1.10}$$

[1] If $g(x)=f(x-\alpha)$ then $\hat{g}(\xi)=\hat{f}(\xi)e^{-2\pi i\alpha\xi}$

Now, we use the change of variable $z = |\xi| y$ and obtain that

$$J(\xi) := \int_{\mathbb{R}^n} \frac{1-\cos(2\pi\xi\cdot y)}{|y|^{n+2s}} dy = |\xi|^{2s} \int_{\mathbb{R}^n} \frac{1-\cos\left(\frac{2\pi\xi}{|\xi|}\cdot z\right)}{|z|^{n+2s}} dz.$$

Now, we consider a rotation R for which $Re_1 = \frac{\xi}{|\xi|}$ and we denote by R^T its transpose. Then, by using the change of variables $\omega = R^T z$ we obtain

$$\begin{aligned} J(\xi) &= |\xi|^{2s} \int_{\mathbb{R}^n} \frac{1-\cos\left(2\pi(Re_1\cdot z)\right)}{|z|^{n+2s}} dz \\ &= |\xi|^{2s} \int_{\mathbb{R}^n} \frac{1-\cos\left(2\pi(e_1\cdot(R^T z))\right)}{|R^T z|^{n+2s}} dz \\ &= |\xi|^{2s} \int_{\mathbb{R}^n} \frac{1-\cos\left(2\pi\omega_1\right)}{|\omega|^{n+2s}} d\omega. \end{aligned}$$

The second equality follows from the fact that if $Re_1 = \frac{z}{|z|}$, by applying R^T (which is equal to R^{-1} since R is a rotation) we obtain

$$R^T R e_1 = \frac{1}{|z|} R^T z \Rightarrow |z| e_1 = R^T z \Rightarrow |z| = |R^T z|.$$

Changing variables $\tilde{\omega} = 2\pi\omega$ (we still write ω as a variable of integration) we obtain that

$$J(\xi) = (2\pi|\xi|)^{2s} \int_{\mathbb{R}^n} \frac{1-\cos\omega_1}{|\omega|^{n+2s}} d\omega. \tag{1.11}$$

Notice that this latter integral is finite. Indeed, integrating outside the unit ball B_1 (see (1.4)) we have that

$$\int_{\mathbb{R}^n\setminus B_1} \frac{|1-\cos\omega_1|}{|\omega|^{n+2s}} d\omega \le \int_{\mathbb{R}^n\setminus B_1} \frac{2}{|\omega|^{n+2s}} d\omega < \infty,$$

while inside the ball we can use the Taylor expansion of the cosine function, by (1.3) and the fact that $s < 1$ we have that

$$\int_{B_1} \frac{|1-\cos\omega_1|}{|\omega|^{n+2s}} d\omega \le \int_{B_1} \frac{|\omega|^2}{|\omega|^{n+2s}} d\omega \le \int_{B_1} \frac{1}{|\omega|^{n+2s-2}} < \infty.$$

Therefore, by taking

$$C(n,s) := \left(\int_{\mathbb{R}^n} \frac{1-\cos\omega_1}{|\omega|^{n+2s}} d\omega\right)^{-1}, \tag{1.12}$$

it follows from (1.11) that

$$J(\xi) = \frac{(2\pi|\xi|)^{2s}}{C(n,s)}. \tag{1.13}$$

By inserting this into (1.10) we obtain that

$$\mathcal{F}\left((-\Delta)^s u(x)\right) = C(n,s)\hat{u}(\xi)J(\xi) = (2\pi|\xi|)^{2s}\hat{u}(\xi).$$

This proves the result by applying the inverse of the Fourier transform.

□

By using the inverse Fourier transform, one has that

$$\begin{aligned} -\Delta u(x) &= -\Delta\left(\mathcal{F}^{-1}(\hat{u})\right)(x) = -\Delta \int_{\mathbb{R}^n} \hat{u}(\xi) e^{2\pi i x\cdot\xi} d\xi \\ &= \int_{\mathbb{R}^n} (2\pi|\xi|)^2 \hat{u}(\xi) e^{2\pi i x\cdot\xi} d\xi = \mathcal{F}^{-1}((2\pi|\xi|)^2\hat{u}(\xi)). \end{aligned}$$

Thus from Proposition 1.2 follows that the classical Laplacian is the limit case of the fraction one, namely for any $u \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\lim_{s\to 1}(-\Delta)^s u(x) = -\Delta u(x), \qquad \lim_{s\to 0}(-\Delta)^s u(x) = u(x).$$

Proposition 1.3. *Let $s \in (0,1)$. The fractional Sobolev space $W^{s,2}(\mathbb{R}^n)$ coincides with $\hat{H}^s(\mathbb{R}^n)$, defined in* (1.2)*. In particular for any $u \in W^{s,2}(\mathbb{R}^n)$ we have*

$$[u]^2_{H^s(\mathbb{R}^n)} = 2C(n,s)^{-1}\int_{\mathbb{R}^n}(2\pi|\xi|)^{2s}|\mathcal{F}u(\xi)|^2 d\xi, \tag{1.14}$$

where $C(n,s)$ is defined in (1.8)*.*

Proof. Firs we prove (1.14). For every fixed $y \in \mathbb{R}^n$, by a changing of variable choosing $z = x - y$ we get

$$\begin{aligned}
\int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^n}\frac{|u(x)-u(y)|^2}{|x-y|^{n+2s}}dx\right)dy &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\frac{|u(z+y)-u(y)|^2}{|z|^{n+2s}}dzdy \\
&= \int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^n}\left|\frac{u(z+y)-u(y)}{|z|^{n/2+s}}\right|^2 dy\right)dz \\
&= \int_{\mathbb{R}^n}\left\|\frac{u(z+.)-u(.)}{|z|^{n/2+s}}\right\|^2_{L^2(\mathbb{R}^n)}dz \\
&= \int_{\mathbb{R}^n}\left\|\mathcal{F}\left(\frac{u(z+.)-u(.)}{|z|^{n/2+s}}\right)\right\|^2_{L^2(\mathbb{R}^n)}dz,
\end{aligned}$$

where the isometry of the Fourier transform in L^2 has been used. Now, using (1.13) we obtain

$$\begin{aligned}
\int_{\mathbb{R}^n}\left\|\mathcal{F}\left(\frac{u(z+.)-u(.)}{|z|^{n/2+s}}\right)\right\|^2_{L^2(\mathbb{R}^n)}dz &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\frac{|e^{2\pi i\xi\cdot z}-1|^2}{|z|^{n+2s}}|\mathcal{F}(u)(\xi)|^2 d\xi dz \\
&= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n}\frac{(1-\cos(2\pi\xi\cdot z))^2+\sin^2(2\pi\xi\cdot z)}{|z|^{n+2s}}|\mathcal{F}(u)(\xi)|^2 d\xi dz \\
&= 2\int_{\mathbb{R}^n}\left(\int_{\mathbb{R}^n}\frac{(1-\cos(2\pi\xi\cdot z))}{|z|^{n+2s}}dz\right)|\mathcal{F}u(\xi)|^2 d\xi \\
&= 2C(n,s)^{-1}\int_{\mathbb{R}^n}(2\pi|\xi|)^{2s}|\mathcal{F}u(\xi)|^2 d\xi.
\end{aligned}$$

This proves (1.14). Finally we prove that the spaces $H^s(\mathbb{R}^n)$, $\widehat{H}^s(\mathbb{R}^n)$ coincide. We prove it by double inclusion. We start supposing that $u \in H^s(\mathbb{R}^n)$ by (1.14) and the isometry of the Fourier transform in L^2 we have

$$\begin{aligned}
\int_{\mathbb{R}^n}(1+|\xi|^{2s})|\mathcal{F}u(\xi)|^2 d\xi &= \int_{\mathbb{R}^n}|\mathcal{F}u(\xi)|^2 d\xi + \int_{\mathbb{R}^n}|\xi|^{2s}|\mathcal{F}u(\xi)|^2 d\xi \\
&= \int_{\mathbb{R}^n}|u(x)|^2 dx + \frac{C(n,s)}{(2\pi)^{2s}}[u]_{H^s(\mathbb{R}^n)} < \infty.
\end{aligned}$$

The last inequality follows also from the fact $u \in L^2(\mathbb{R}^n)$. This proves that $u \in \widehat{H}^s(\mathbb{R}^n)$. Now, we suppose that $u \in \widehat{H}^s(\mathbb{R}^n)$. By (1.14) we have

$$\begin{aligned}
[u]_{H^s(\mathbb{R}^n)} &= 2(2\pi)^{2s}C(n,s)^{-1}\int_{\mathbb{R}^n}(2\pi|\xi|)^{2s}|\mathcal{F}u(\xi)|^2 d\xi \\
&\leq 2C(n,s)^{-1}\int_{\mathbb{R}^n}(1+|\xi|^{2s})|\mathcal{F}u(\xi)|^2 d\xi < \infty
\end{aligned}$$

This proves that $u \in H^s(\mathbb{R}^n)$.

$\square$

Finally, we are able to prove the relation between the fractional Laplacian operator $(-\Delta)^s$ and the fractional Sobolev space $H^s(\mathbb{R}^n)$.

Proposition 1.4. *Let $s \in (0,1)$ and let $u \in H^s(\mathbb{R}^n)$. Then*

$$[u]^2_{H^s(\mathbb{R}^n)} = 2C(n,s)^{-1}\|(-\Delta)^{\frac{s}{2}}u\|^2_{L^2(\mathbb{R}^n)}, \tag{1.15}$$

where $C(n,s)$ is defined by (1.8).

Proof. The equality (1.15) follows from Proposition 1.2 and Proposition 1.3. Indeed we have

$$\begin{aligned}
\|(-\Delta)^{\frac{s}{2}}u\|_{L^2(\mathbb{R}^n)} &= \|\mathcal{F}(-\Delta)^{\frac{s}{2}}u\|^2_{L^2(\mathbb{R}^n)} \\
&= \|(2\pi|\xi|)^s\mathcal{F}u\|^2_{L^2(\mathbb{R}^n)} \\
&= \frac{1}{2}C(n,s)[u]^2_{H^s(\mathbb{R}^n)}.
\end{aligned}$$

$\square$